

INFERTILITY

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Introduction

- Infertility is a global health issue affecting millions of people of reproductive age worldwide
- WHO estimated that between 48 million couples and 186 million people live with infertility globally
- It is estimated to affect between 8 and 12% of couples of reproductive age worldwide
- It is a significant psychological, social and medical problem affecting couples worldwide

Cont...

- In sub-Saharan Africa the incidence rises to 20-30% the assumption is in view of the high prevalence of STDs, induced abortion, deliveries under unhygienic conditions and TB
- In Ethiopia, The overall prevalence of infertility is 27.6%
- Of these, 14.4% had primary infertility, and 13.2% had secondary infertility
- Sterility affects 1–2% of couples

Cont...

➤ Definition

- **Infertility** is defined as the inability to conceive after 12 or more months of regular intercourse of reasonable frequency without contraception
- It may be a disease of the **male or female** reproductive system
- **Infertility** implies a decrease in the ability to conceive
- **Sterility** implies an intrinsic inability to achieve pregnancy
- **Fecundability** is expressed as the likelihood of **conception per month** of exposure
- **Fecundity** is the probability of **achieving a live birth in 1 menstrual cycle**, the monthly probability of conceiving is 20 to 25 %.

Cont...

➤ Types of infertility

Primary infertility

- ✓ Couple has never produced a pregnancy

Secondary infertility

- ✓ Women has previously been pregnant ,regardless of the outcome and now unable to conceive

Etiology of infertility

- In 85-90% of infertile couples the etiology is found
- Male cause only 25-40%
- Female cause only 40-55%
- Both 10%
- 10-15% the infertility is unexplained.

Etiology of male infertility

- ✓ The causes of male infertility can be divided into four main areas:

Etiology of infertility	Site of dysfunction	Prevalence %
Pre-testicular	Hypothalamic / pituitary	1 -2
Testicular	Testis	30 - 40
Post-testicular defects	Ducts of sperm transport	10 - 20
Idiopathic	-----	40 -50

1. Pre-testicular causes

Are hypothalamic & pituitary disease

A. Congenital disorders

- Endocrine → hypogonadism
- Kallmann's syndrome
- **Erectile & Ejaculatory failure**

B. Acquired : Tumors

- Infiltrative diseases
- **Erectile & Ejaculatory failure**
- Chronic disease

- **Hyperprolactinemia & Drugs**

Cont...

2. Testicular Causes

A. Congenital disorders

- Klinefelter's syndrome
- Y chromosome defects
- Cryptorchidism & Varicoceles
- Torsion

B. Acquired disorders

- Infection
- Drugs
- Radiation
- Environmental factors
- Smoking & Systemic disorders

Cont...

3. Post-testicular causes

- Congenital or acquired disease in the ductal system of a male
- Absence, dysfunction, or obstruction of the epididymis
- Bilateral obstruction, ligation, or altered peristalsis of the vas deferens
- Postvasectomy

4. Idiopathic male infertility

- The cause of abnormal sperm number, morphology or function cannot be identified Despite careful assessment of all possible causal mechanisms
- men who have repeatedly normal semen analyses but cannot impregnate an apparently normal female partner.

Diagnosis of male infertility

➤ History

- Type of infertility (primary or secondary)
- history of impotence
- premature ejaculation
- change in libido
- history of testicular trauma
- Infections

➤ Diagnostic work up of male infertility

1. Standard semen analysis

- Cornerstone of the assessment of the male infertility
- The standard semen analysis consists of the following:
 - Measurement of semen volume and pH
 - Microscopy for debris and agglutination
 - Assessment of sperm concentration, motility, and morphology
 - Sperm leukocyte count
 - Search for immature germ cells
 - A normal semen analysis excludes male factor 90% of the time

Normal semen analysis findings

Cont...

Volume	>1.5 mL
Count	>20 million/mL
Motility	>50%
Morphology	>30%
WBCs	<1 million/mL
Total Sperm Number	>40 million/ejaculate
Morphology	≥ 30% with normal forms
Viscosity	Liquification in 30-60 min
Vitality	≥ 75% live
Sperm concentration	>20 million/ml
PH	7.2-7.8

Cont...

- **Aspermia:** failure of formation or emission of semen/no ejaculate
- **Azoospermia:** no spermatozoa in the ejaculate
- **Oligospermia:** sperm concentration fewer than 20 million/ml
- **Polyzoospermia:** >350 million/ml
- **Asthenozoospermia:** abnormal motility
- **Teratozoospermia:** fewer than 30% spermatozoa with normal morphology
- **Necrozoospermia:** dead or motionless
- **Oligoasthenoteratozoospermia** - disturbance of all three variables.

Cont...

2. Sperm Function Tests

a. Sperm Penetration Assay

b. Human Zona Binding Assay

3. Endocrine Evaluation

4. Genetic Evaluation

5. Ultrasonography

- Scrotal Ultrasonography
- Transrectal Ultrasonography (TRUS)

Etiology of female infertility

- Ovulatory Dysfunction
- Elevated prolactin
- Early menopause
- Tubal /Peritoneal Factor
- Endometriosis
- Uterine Factor
- Cervical Factors
- Vaginal Factors
- Systemic diseases
- personal habit & Unexplained

Diagnosis

➤ History

- Type of infertility (primary or secondary)
- Detailed history of previous pregnancies
- Specific questions
- Frequency of intercourse
- Use of lubricants
- menstrual history
- frequency
- patterns since menarche

Cont...

➤ Diagnostic work up of female infertility

❖ Detection of ovulation

A. Indirect

1. Evaluation of peripheral or end organ change

i. BBT

ii. Cervical mucus study

iii. Vaginal cytology

2. Hormone estimation

3. Endometrial biopsy – secretory changes

Subnuclear vacuolation 36-48 hr after ovulation

Cont...

B. Direct

- Laparoscopy
- Transvaginal Ultrasonography (TVUS)
- Hysterosalpingography (HSG)
- Hysteroscopy
- Pelvic ultrasound

❖ Anovulation / oligo –ovulation confirmed

- i. Serum FSH
- ii. Prolactin
- iii. TSH
- iv. Serum Dehydroepiandrosterone (DHEA), Testosterone

Managements of Infertile couple

- I. Lifestyle Therapies
- II. Correction of an Identified Cause
- III. Assisted Reproductive Technologies

Cont...

I. Lifestyle Therapies

- Weight Optimization
- Stop Smoking
- Exercise
- Stress Management

Cont...

II. Correction of an Identified Cause

A. Ovulation Induction for ovulatory dysfunction

Drugs used for ovulation induction

❖ **Clomiphene citrate:** The most common medications used to stimulate ovulation

- Clomiphene is a medication patients take by mouth (orally)
- It causes the body to make more of the hormones that cause the eggs to mature in the ovaries

❖ **hMG**

❖ **FSH**

❖ **hcG**

❖ **GnRH & GnRH analogs**

Cont...

B. Surgical correction of tubal of infertility to;

- to remove patches of endometriosis, uterine fibroids, scarring and repair the tubes or remove blockages in the tubes
- Adhesion lysis, polypectomy, myomectomy
- Reversal of prior surgical sterilization
- Surgical treatment in some cases (varicocele)

Cont...

III. Assisted Reproductive Technologies

- ✓ used in infertile couples for whom direct corrections of underlying causes are not feasible
- ✓ These technologies are still largely **unavailable, inaccessible and unaffordable** in many parts of the world, particularly in low and middle-income countries like Ethiopia
- ✓ These techniques include,
 1. **Artificial/ intrauterine insemination**: Intrauterine insemination can be performed either with patient's or donor's sperm
 - *Donor semen should be free from STDs/HIV*

Cont...

2. In vitro fertilization (IVF): Involves retrieving eggs and sperm from female and male partners and placing them in a lab dish to enhance fertilization

- Fertilized eggs are transferred several days later into the uterus

3. Intracytoplasmic sperm injection (ICSI): Involves injection of single sperm into the egg

- The woman is administered fertility drugs prior to the procedure to aid in the production of multiple eggs
- Only active undamaged sperm are selected for injections

Cont...

4. Egg donation

5. Gestational carrier surrogacy

6. Gamete intrafallopian transfer (GIFT): placing a mixture of sperm and eggs directly into the woman's fallopian tube

- GIFT does **not** allow visual confirmation of fertilization

7. zygote intrafallopian transfer (ZIFT): ZIFT is also called tubal embryo transfer, is another variation of IVF

- As with IVF, the actual fertilization takes place in a lab dish
- Fertilized eggs are placed directly into a fallopian tube

Prevention of Infertility in women

- Take steps to prevent sexually transmitted diseases
- Avoid illicit drugs
- Adopt good personal hygiene and health practices
- Have annual check ups with your GYN once you are sexually active
- Regular exercise
- Avoid alcohol, tobacco, caffeine, and narcotics
- Limit medications
- Eat a balanced diet

Genito-urinary Complications

Pelvic Organ Prolapse(POP)

- **POP** is the downward descent of the uterus and the vagina from their normal pelvic positions or **the herniation of the pelvic organs to or beyond the vaginal walls,**
- Many women with prolapse experience symptoms that impact daily activities, sexual function, and exercise
- The presence of POP can have a detrimental impact on body image and sexuality
- Accounts for 13% of hysterectomy
- It is a Common condition in post menopausal women

Risk factors of POP

❖ Obstetric cause

- Multiparty
- Instrumental delivery
- Mismanagement of third stage of labor
- Unnecessary use of fundal pressure
- Childbirth related trauma to the pelvic support and improperly repaired genital tract tears
- Infant birth weight >4,500 g
- Forceps delivery
- Prolonged second stage of labor

Cont...

❖ Chronic disease

- DM
- Connective tissue disorders
- Hypertension & Coughing (chronic TB)

❖ **Aging:** Estrogen deficiency leads to slow weakening of both pelvic floor and loss of elasticity and strength of structures

❖ Conditions w/c increase intraabdominal pressure

- Strenuous exercise
- Obesity
- Constipation
- Chronic cough
- pelvic tumor and ascitis

Classification of POP

- Fundamentally, there are two types of genital prolapse

I. Vaginal prolapse

A. Anterior vaginal wall prolapse

- **Cystocele**- herniation of the Bladder
- **Urethrocele**: herniation of the posterior wall of urethra into the lower part of the vagina
- **Cysto-urethrocele**: herniation of the entire anterior vaginal wall

B. Posterior vaginal wall prolapse

- **Rectocele**: herniation of the rectum into the vagina,
- **Enterocele**- herniation of the of the intestines to or through the vaginal wall
- **Vault prolapse** (upper portion of the vagina): refers to a herniation of a peritoneal sac at the vaginal vault after abdominal or vaginal hysterectomy

Cont...

II. Uterovaginal prolapse

- A down ward descent of **the uterus and vagina** from their normal pelvic position
- Is measured in relation to degree of prolapse of the cervix below the level of **ischial spine** (the normal station of the cervix in the pelvis)
- The uterine component of the uterovaginal prolapse is measured in relation to the degree of prolapse of the cervix below the level of the **ischial spines**

Clinical manifestations of POP

- The symptoms of genital prolapse could be **general or specific**.

❖ **general symptoms**

- **Painless mass bulging** through the introitus which is prominent when standing or straining
- **Pelvic pressure or bearing down sensation** in the pelvis or sensation of something coming down or dragging discomfort and low back pain
- **Urinary symptoms** (a sensation of incomplete emptying and, in rare cases, complete urinary retention).
- **Defecatory symptoms** (constipation, fecal urgency, fecal incontinence and obstructive symptoms).

Cont...

❖ Specific symptoms

- **For anterior vaginal wall prolapse:** stress incontinence, symptoms of urinary tract infection, sense of incomplete emptying and urinary retention
- **For posterior vaginal wall prolapse:** constipation, rectal fullness, difficulty in emptying the bowel, sense of incomplete emptying which necessitates splinting or digital removal
- **For enterocele:** usually nonspecific symptoms resulting from traction of the abdominal viscera
- **For uterine prolapse:** abnormal uterine bleeding, abnormal vaginal discharge

Diagnosis of POP

❖ History

- Bilateral groin pain
- Back ache
- Coital difficulty
- Hx of prolonged labor
- Hx of instrumental delivery
- Hx of retained placenta
- A medical history is also important to elicit prolapse-associated symptoms
- Hx of chronic diseases

Cont...

❖ P/E

- Nutritional status (obese or not)
- Mass (reducible/not, mobile/not, ulceration, color and location)
- visualization or palpation of the cervix with or without the patient straining and palpation of the uterus during pelvic examination

Grading/staging/degrees of Uterovaginal prolapse

➤ Three systems

A. Degree system: three degrees(RP=V/introitus)

Grade I: Descent cervix below ischial spines but not beyond introitus

Grade II: Descent cervix beyond introitus, but not all uterus (fundus is inside)

Grade III: The whole uterus is outside the introitus

Grade IV: Complete eversion, not replaced manually, keratinized, edematous, infected some times called proidentia

Cont...

B. Baden-Walker half way system(RP=Hymen):

1. **Grade 0:** normal position
2. **Grade 1:** Halfway between ischial spine to hymen
3. **Grade 2:** In to hymen
4. **Grade 3:** halfway past hymen to maximal descent
5. **Grade 4:** Maximal possible descent past hymen for each site

Cont...

C. POPQ system(RP=Hymenal ring)

- The POPQ system is an objective, site-specific system for describing and staging POP in women.
- The POPQ system involves quantitative measurements of various points representing anterior, apical, and posterior vaginal prolapse to create a "topographic" map of the vagina
- The POPQ has proven interobserver and intraobserver reliability.

Cont...

- **Has four stages**

1. **Stage 0:** no prolapse
2. **Stage 1:** maximal descent is 1cm above hymenal ring
3. **Stage 2:** up to 1cm beyond the hymenal ring
4. **Stage 3:** up to 2cm from total vaginal length from hymenal ring
5. **Stage 4:** beyond the hymenal ring to the extent of total vaginal length

KEY:RP=Reference Point

Cont...

Grade-IV UVP **Proidentia**



Figure 24.5 Proidentia of the uterus and vagina.



Figure 24.4 Uterine prolapse with apical detachment from the uterosacral ligament complex and lateral wall detachment from the endopelvic connective tissue.

Approach to management

- Treatment of POP requires significant *health care resources*

➤ Indications for treatment:

- women with symptoms of prolapse or associated conditions (urinary, bowel, or sexual dysfunction).
- Obstructed urination or defecation or hydronephrosis from chronic ureteral kinking are all indications for treatment, regardless of degree of prolapse

Cont ...

- Women with *symptomatic prolapse* can be managed expectantly, or treated with conservative or surgical therapy.

❖ **Expectant management** : Expectant management is a viable option for women who can tolerate their symptoms and prefer to avoid treatment

❖ **Conservative therapy**: is the first line option for all women with POP

- Vaginal pessary placed in the vagina that support prolapse
- Estrogen therapy for postmenopausal women
- Pelvic floor muscle((Kegel) exercises)exercises

Cont...

❖ **Surgical treatment**

- Surgical candidates include women with symptomatic prolapse who have failed or declined conservative management of their prolapse.
- Surgery for POP aims to relieve symptoms and to restore normal anatomic relationships.
- The surgical procedure and approach (abdominal or vaginal) is tailored to the particular type of POP present.

Cont...

- Hysterectomy for uterine prolapse
- Anterior repair, paravaginal repair for cystocele
- Posterior repair for rectocele
- Enterocele repair
- Vaginal vault suspension (sacrospinous suspension)
- Perineorrhaphy for relaxed vaginal outlet.

Complications of POP

- Keratinization of the vagina and cervix
- Decubitus ulcer on the cervix
- Urinary tract infection and urinary obstruction (angulation of the urethra or constriction of the ureters)
- Sexual dysfunction

Prevention of POP

- weight loss
- avoidance of jobs that require heavy lifting
- Encouraging postnatal perineal exercises
- Proper suturing of perineal and vaginal lacerations with proper anatomic restoration
- Avoidance of excessive fundal pressure during delivery
- Treatment of chronic cough and constipation

GENITO URINARY FISTULA

- **Fistula:** is an abnormal communication between two hollow organs or two epithelial surfaces
- **Genital fistula:** is Abnormal communication between **the genital system** (uterus, cervix, and vagina) **and the urinary system** with the urinary tract (urethra, bladder, **and** ureter) and/or **lower intestinal tract** (rectum, anus`)
- **Obstetric fistula:** a genital fistula occurring as a result of obstetric problems (prolonged & obstructed labor) = > also called “ **a child birth injury**”
- Global estimate is about 100,000 new cases per year.
- Its **common** cause (90%) in **developing** countries
- **54-60 %** of them were divorced

Classification

- Method of classification is based on **anatomic position of the communication**
- Fistulas can develop **at any** point between the **genital** and **urinary** systems
- **Vesicovaginal fistula (VVF)**: is communication between the bladder and the vagina
 - **VVF** is the **most common** among others
- **Recto vaginal fistula(RVF)** :) RVF is an abnormal connection between the **rectum and the vagina**. This allows fecal material to pass freely from the rectum to the vagina
- **Both VVF & RVF**: (Double fistula):The incidence of combined fistulae ranges from 5% to 10%

Cont...

- **Vesicouterine (VUF):** communication between the uterus and the bladder
- **Uretero Uterine Fistula(UUF):** communication between the **ureter** and the **uterus**
- **Ureterovaginal fistula:**communication between the **ureter** and **vagina**
- **Urethrovaginal fistula:**communication between the **urethra** and **vagina**

Types of Fistula

Bladder

Vesicovaginal
Vesicourthrovaginal
Vesicouterine
Vesicocervical

Urethra

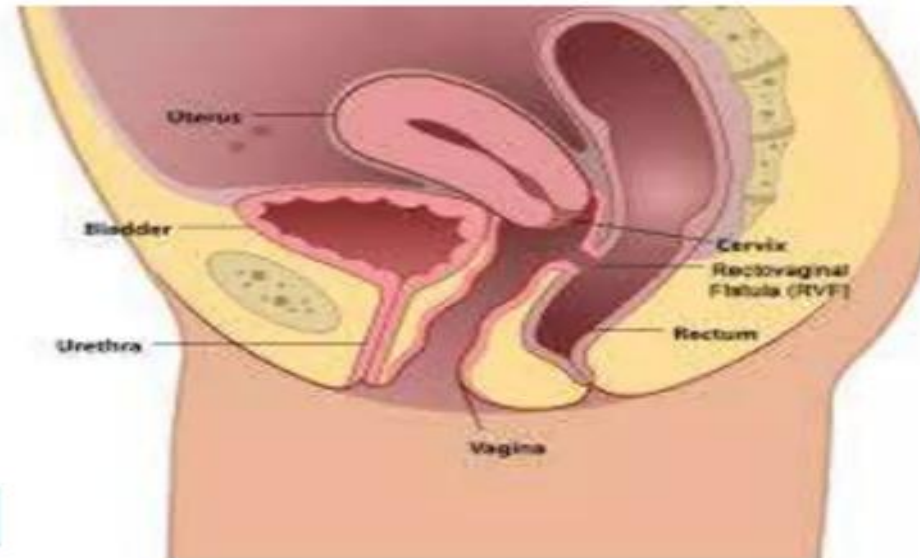
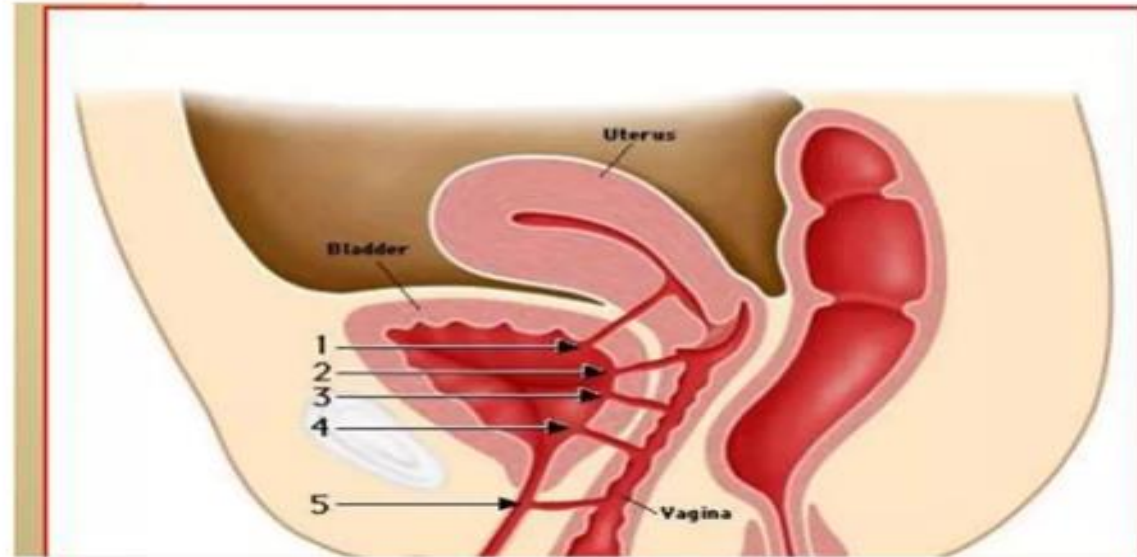
Urethrovaginal

Ureter

Ureterovaginal
Ureterouterine
Ureterocervical

Rectum

Rectovaginal



Etiology

➤ **Obstetrics Causes:** Obstructed labour :**the most common**

- ✓ Traumatic Instrumental delivery
- ✓ Pelvic Surgery (hysterectomy ,c/s, Symphysiotomy)
- ✓ Destructive delivery

➤ **Non-Obstetric Causes :** Penetrating trauma.

- ✓ Female genital mutilation
- ✓ Early marriage & Sexual assault (rape)

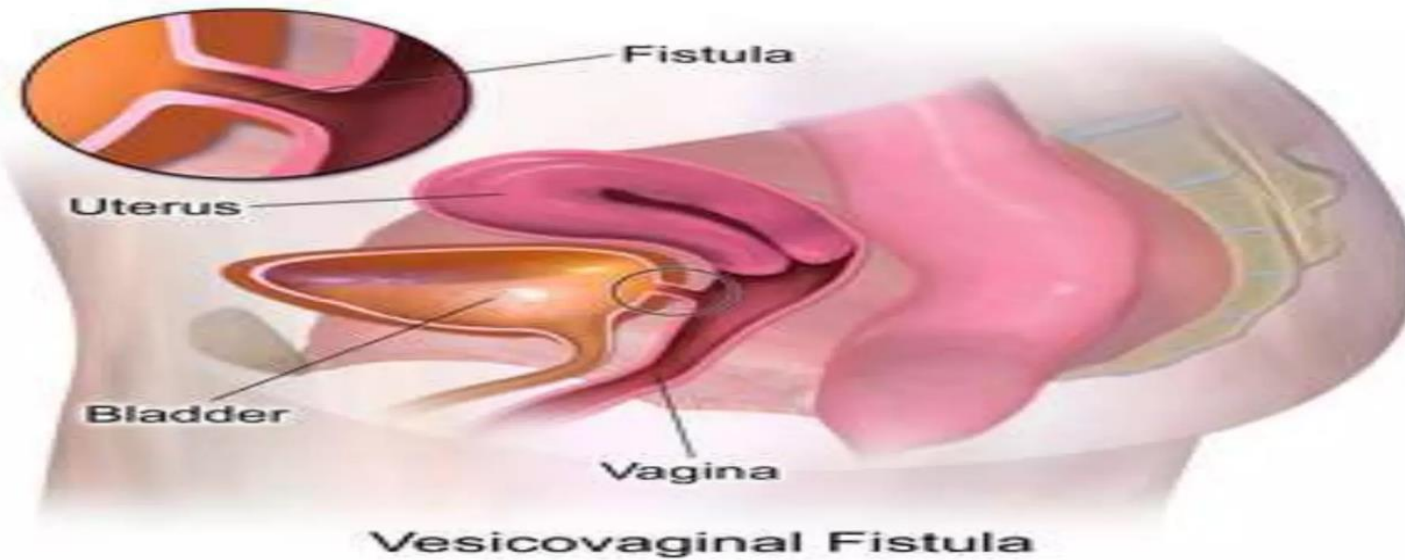
➤ **Others:** Radiotherapy

- ✓ Infection
- ✓ Malignancy & Congenital malformation

Clinical feature of VVF

- Patient presents with leakage of excessive volume of fluid per vagina with no voided urine
- Have foul ammoniac odor
- Vulvar excoriation / ammoniac dermatitis
- Depression / Insomnia
- Disrupted sexual relationship
- Low self esteem
- Foot drop

Cont...

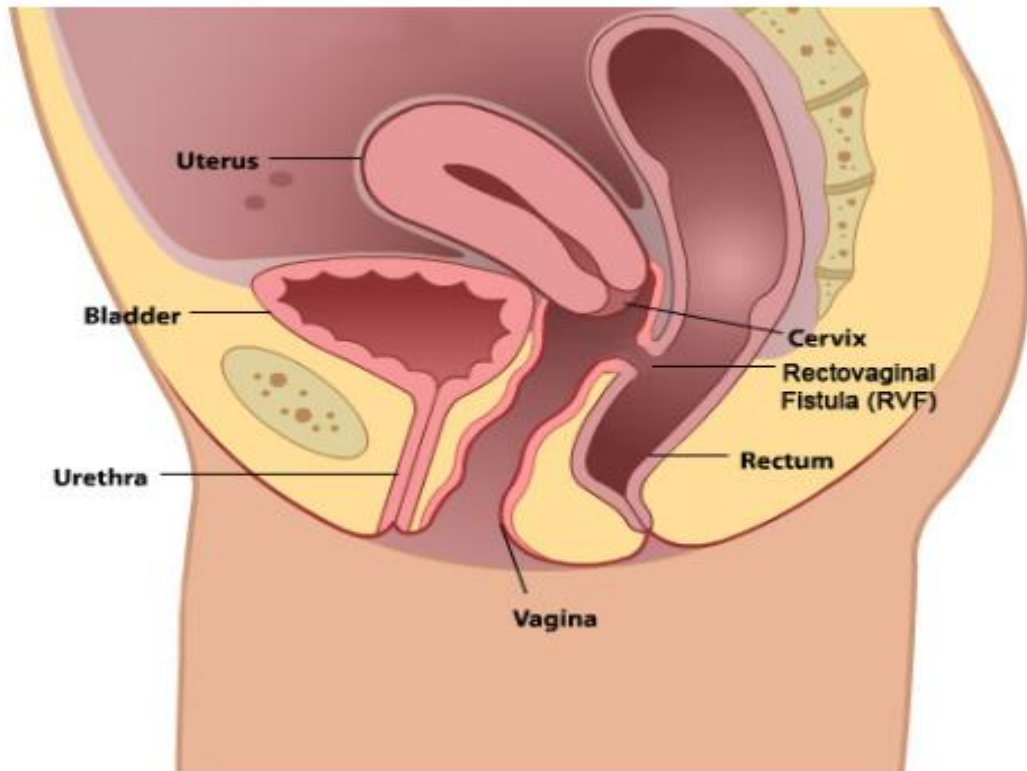


Vesico-vaginal fistula

Dr.SMRUTI RANJAN PATANAİK
GUIDE – DR . NINA MISHRA

Clinical features of RVF

- continuous leakage of stool
- gas per vaginum



- **Ureterovaginal fistula:** Small amount of urine leakage via vagina with regular voiding
- **Urethrovaginal fistula:** Leakage during and after voiding via vagina

Diagnosis of Genital fistula

➤ History

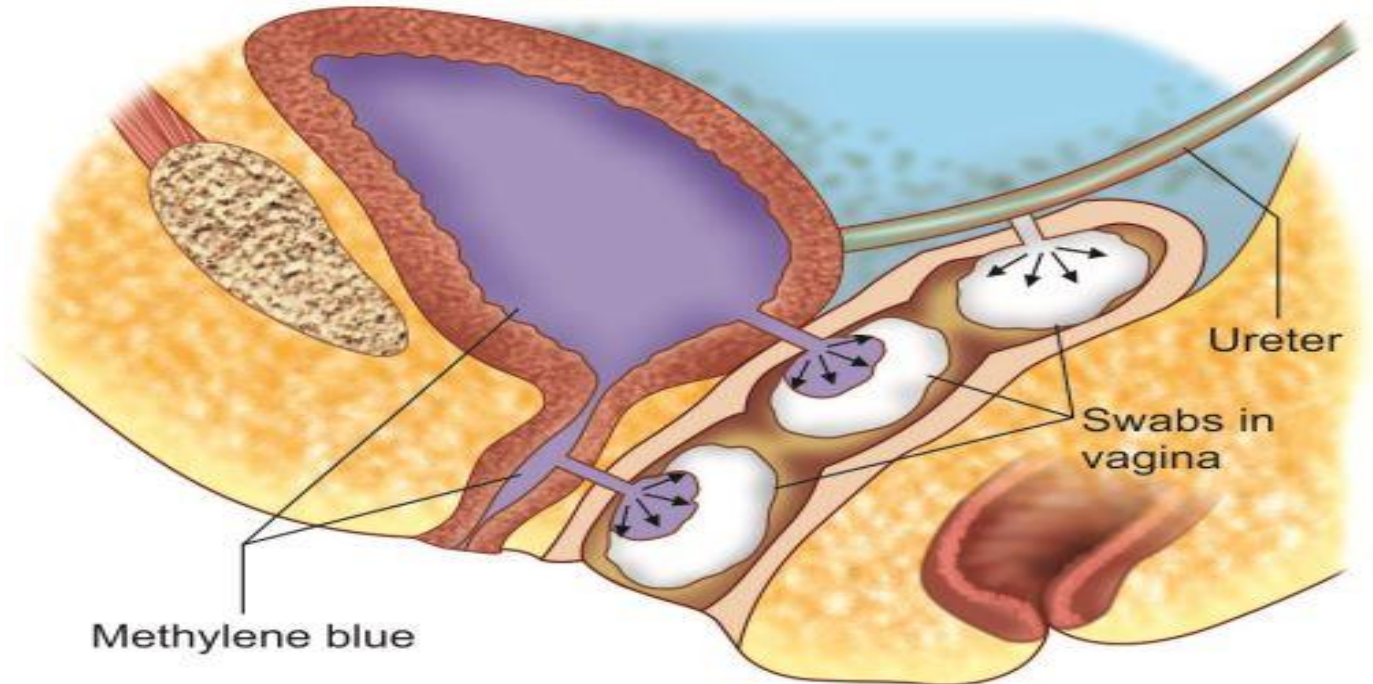
➤ Investigation

- Vaginal fluid analysis
- Urine analysis
- Urine culture
- **Dye test (3-swab test)**
- RFT
- RBS
- PICT

Diagnosis of Genital fistula

Dye test (*Three-swab test*)

- **Three** separate sponge swabs, one above the other, are placed in the **upper, middle, and lower vagina**
- **The bladder is then filled** with 300 ml methylene blue through Foley catheter
- the catheter is then removed and the patient will be asked to walk around for half an hour, but not pass urine

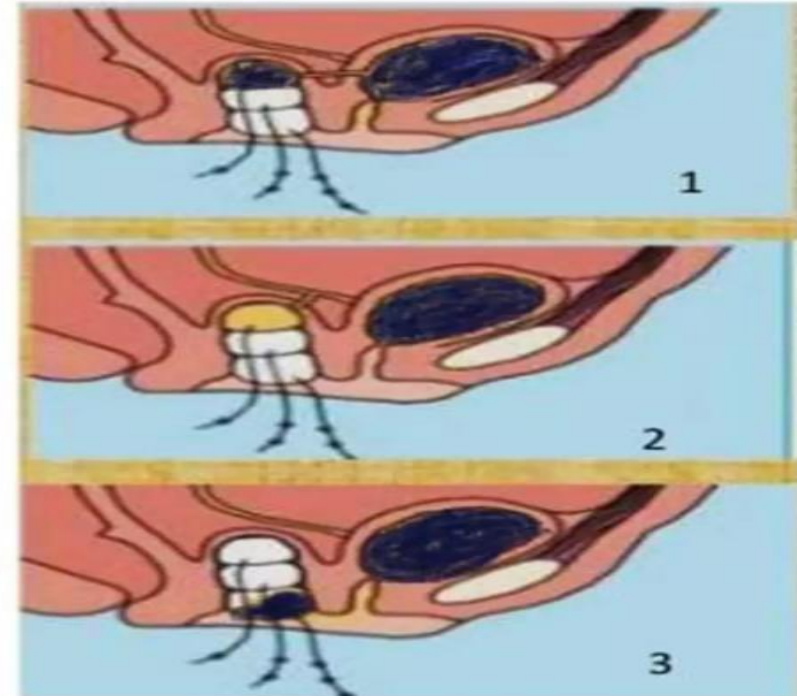


■ The specific gauze coloured with dye suggests where in the vagina a fistulous tract is located

Three swab test

Result of 3 swab test

1. Discolouration of topmost or middle swab →
vesicovaginal fistula
2. Uppermost swab wetting but not discolouration
→ **Ureterovaginal fistula**
3. Discolouration of lower most swab but upper two swabs remain dry
→ **Urethrovaginal fistula**



Management of Genital Fistula

➤ Conservative Management

■ Bladder catheterization

- ✓ Indicated for Fistula diagnosed early after formation
- ✓ Small fistula
- ✓ Duration of catheterization: 2 -4 weeks

■ Advantage:

- Continuous drainage
- Keeps bladder at rest

Cont...

➤ Surgical Management

✓ Pre operative care:

- Schedule After **three months** for healing of the necrotized tissues around the hole
- Treat any infection or inflammation
- Correct nutritional status
- Perineal skin care
- Physiotherapy for patients with foot drop & joint contractures
- Colostomy for patients with high RVF

Cont...

✓ **Surgery:** It is definitive treatment, vaginal approach

✓ **Vaginal approach:** Safer and convenient to the patient

✓ 98-100% success rate

✓ **Abdominal approach:** Large fistula, Fistula adjacent to ureters or involving ureters e.g., ureteric reimplantations

✓ Success rate 71-100 %

Strategies to prevent Obstetric fistula

➤ Primary Prevention Strategies

- Improve nutrition
- Ensure immunization and treatment of childhood diseases
- Improve access to information and service for all components of reproductive health services
- Educate to avoid early marriage
- Prevent harmful traditional practices
- Empower women
- Ensure availability and use of family planning

Cont...

➤ Secondary Prevention Strategies

- Ensure access to and use of antenatal care
- Use partograph during labor
- Ensure presence of skilled attendant at delivery
- Improve accessibility and availability of 24- hour emergency obstetric care
- Improve the health care delivery system

Cont...

➤ Tertiary Prevention Strategies

- Catheterize in cases of obstructed labor and small fistula
- Provide supportive care
- Ensure access to and availability of surgical repair of fistula

Psychosocial Consequences of Fistula

❖ Social isolations from;

- Churches
- mosques
- market places
- Wedding & funerals

❖ Divorce

❖ Depression

❖ Psychosis

❖ Suicide

Reading assignment

- **Types of urinary incontinence**
- **Causes of urinary incontinence**
- **Clinical Features and diagnosis of urinary incontinence**
- **Nursing assessment, diagnoses, and intervention for patients with urinary incontinence**

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THANK YOU!